

Tarkov Weapon Mod Optimizer

Automated Build Optimization using Constraint Programming

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The Problem: Complexity in Escape from Tarkov

Escape from Tarkov features one of the most complex weapon modification systems in gaming.

Challenges for New Players:

- **Overwhelming scale:** 2,500+ mods, each affecting recoil, ergonomics, weight, accuracy
- **Nested compatibility:** Mods attach to mods (rail → mount → sight)
- **Combinatorial explosion:** A single AR-15 has 50+ slots, millions of valid builds
- **Non-obvious trade-offs:** Is a 50k grip worth it over a 5k one?
- **Dynamic availability:** Prices change; items locked behind trader levels, quests, and flea market access

Result: Players spend hours theory-crafting or blindly copy “meta builds” without understanding trade-offs.

The Solution: Automated Optimization

Project Goal: Create a tool that mathematically guarantees the "best" weapon build for a given set of constraints.

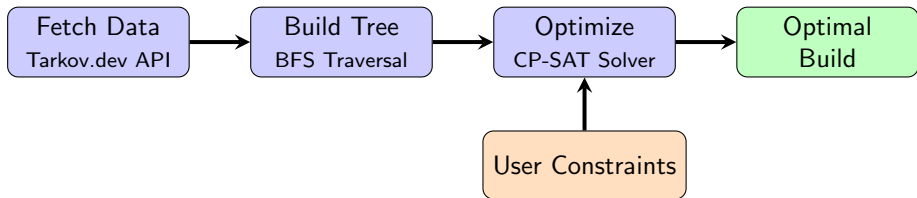
Use Cases:

- *"I have 200k roubles—what's the best M4 I can build?"*
- *"What's the absolute lowest recoil AK, money no object?"*
- *"I'm level 15 with LL2 traders—what can I actually buy?"*
- *"Is spending 100k more actually worth it?"*
- *"Show me all viable builds on a budget curve"*

Key Capabilities:

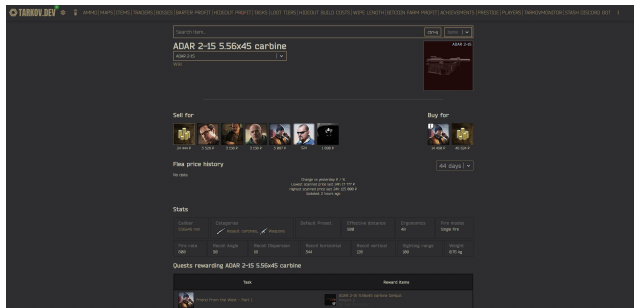
- **Multi-objective:** Balance recoil, ergonomics, and price
- **Budget-aware:** Hard limit or soft penalty on cost
- **Trader filtering:** Respect loyalty levels (LL1–LL4)
- **Flea market toggle:** Include or exclude market prices
- **Pareto frontier:** Visualize price vs. performance trade-offs

Pipeline Overview



Tarkov.dev GraphQL API

- Queries all guns, mods, and compatibility rules
- **Up-to-date prices** from traders and flea market
- Trader loyalty level requirements
- Real-time flea market prices
- Local caching for speed



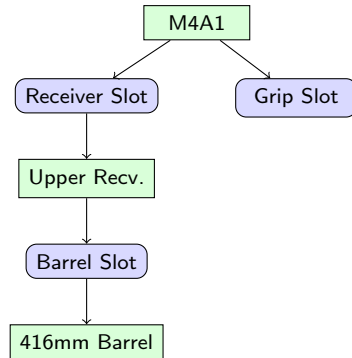
Building the Compatibility Tree

BFS Process:

- 1 Start from the base weapon's slots
- 2 For each slot, find all allowed items
- 3 Add those items to the queue
- 4 For each item, discover *its* slots
- 5 Repeat until no new items found

Data Structures Built:

- `slot_items`: Slot $\rightarrow$ [compatible items]
- `slot_owner`: Slot $\rightarrow$ parent item
- `item_to_slots`: Item $\rightarrow$ [owned slots]



Items own slots; slots contain items.

Introduction to CP-SAT Solver

CP-SAT = **C**onstraint **P**rogramming + **SAT** (Boolean Satisfiability)

What is it?

A solver from Google's OR-Tools library that finds optimal solutions to combinatorial problems by combining:

- **Constraint Programming:** Express complex rules declaratively
- **SAT Solving:** Efficient Boolean logic reasoning
- **Linear Programming:** Optimize numerical objectives

Why CP-SAT for this problem?

- Handles Boolean decisions naturally (select mod or not)
- Scales to thousands of variables and constraints
- Guarantees optimal solution (not just "good enough")
- Solves in milliseconds for typical weapon builds

Applying CP-SAT to Weapon Builds

Core Idea: Define what a valid build looks like, let the solver find the best one.

The Three Components

- 1 **Variables:** Boolean X_i for each reachable mod ($X_i = 1$ if selected)
- 2 **Constraints:** Compatibility rules, budget limits, required slots
- 3 **Objective:** Maximize ergonomics, minimize recoil and price

The solver explores millions of mod combinations, pruning invalid builds instantly, and returns the mathematically optimal configuration.

Constraint 1: Slot Mutex (One Item Per Slot)

Rule: Each slot can hold at most one item.

Formula

For each slot s with candidate items $\{i_1, i_2, \dots, i_n\}$:

$$\sum_{i \in \text{slot}_s} X_i \leq 1$$

where $X_i = 1$ if item i is selected, 0 otherwise.

Example: A pistol grip slot can have a Zenit RK-3 *or* an Ergo PSG-1, but not both simultaneously.

Constraint 2: Parent Dependency

Rule: A mod can only be selected if its parent slot owner is also selected.

Formula

If item c (child) fits in a slot owned by item p (parent):

$$X_c \leq X_p$$

A child cannot be selected without its parent.

Example: You cannot attach a foregrip to a handguard unless that handguard is installed on the weapon.

Gun $\rightarrow$ *Handguard* $\rightarrow$ *Foregrip*

Constraint 3: Conflicting Items

Rule: Some items cannot be used together (API-defined conflicts).

Formula

For each conflict pair (a, b) from the API:

$$X_a + X_b \leq 1$$

At most one of the conflicting items can be selected.

Example: Certain dust covers conflict with specific charging handles due to physical interference—the game prevents mounting both.

Constraint 4: Required Slots

Rule: Certain slots *must* be filled for a functional weapon.

Formula

For each required slot s :

$$\sum_{i \in \text{slot}_s} x_i \geq 1$$

At least one compatible item must fill the slot.

Vital Slots (AR-15 platform):

- Barrel, Gas Block, Pistol Grip, Charging Handle, Receiver, Handguard

Without these, the weapon cannot function in-game.

Constraint 5: Budget Limit

Rule: Total build cost must not exceed the player's budget.

Formula

$$\sum_i \text{Price}_i \cdot X_i \leq \text{Budget}$$

The sum of all selected item prices must stay within budget.

Smart Pricing:

- Considers trader loyalty levels (LL1–LL4)
- Checks flea market availability
- Automatically finds the cheapest source for each item

The Objective Function

Goal: Find the build that best matches the player's priorities.

Weighted Objective (Maximize)

$$\text{Score} = W_E \cdot \text{Ergo} - W_R \cdot \text{Recoil} - W_P \cdot \text{Price}$$

- W_E : Ergonomics weight (higher = faster ADS, less stamina drain)
- W_R : Recoil weight (lower = more controllable)
- W_P : Price weight (lower = cheaper build)

Example Profiles:

- *Meta Build*: $W_R = 100$, $W_E = 0.5$, $W_P = 0$
- *Budget Build*: $W_R = 1$, $W_E = 1$, $W_P = 0.5$

User Interface: Build Optimization

Language
中文

选择武器
选择一把武器:
Custom Guns NL545 (...)

基础属性:

- 人机工效: 48
- 垂直后坐力: 114
- 水平后坐力: 330

可用预设 (1)

玩家与商人

- 玩家等级
79
- 交易市场
- 商人等级
- 约束条件
- +/- 包含/排除模组

Deploy

塔科夫武器模组优化器

使用约束来优化您的武器配置

✓ 已加载 167 把武器和 2181 个模组

探索武器 优化配置 机瞄的任务

优化您的配置

优化权重

人机 已选力 价格 均衡



人机: 34% | 已选力: 33% | 价格: 33%

手动调整权重

优化配置

工作原理: 点击优化配置找到最佳模组搭配

工作原理:

- 使用三角图设置优先级权重 (点击网格点)
- 可设置准确性约束 (预测、最小人机、最大后坐力)
- 点击优化配置找到最佳模组搭配

提示: 先使用探索权重选项卡了解可达到的效果!

Optimization Results: Min Recoil

Language
中文

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可用预设 (1)

玩家与商人

- 玩家等级: 79
- 跳蚤市场
- 商人等级
- 约束条件
- 包含/排除模组

最终属性

人机工效 ①
26.3
+21.7

垂直后坐力 ①
48.3
+45.7

水平后坐力 ①
139.9
+106.1

总重量 ①
4.20 公斤

总配置费用 ①
¥299,267
+¥140,234

已选配置

预览: Custom Guns NL545 (GP) 5.45x39 assault rifle Default
套餐价格: ¥153,033 (RealMarket)
包含: 14 items

额外模组:

	名称	人机	后坐力	价格	来源
	AK-74 "Salga 345" 5.45x39 10-round magazine	+5.0	-	¥1,566	Free Lv25
	AR-15 Strike Industries Advanced Receiver Extension	+2.0	-2.0%	¥10,440	Mechanic LL4
	HK 416 Strike Industries CRUX 15 inch M-LOK handguard	+11.0	-3.0%	¥16,700	Mechanic LL4
	AR-15 Magpul PRS GEN3 stock (Black)	+3.0	-24.0%	¥25,522	Free Lv20
	AR-15 AAC Blackout SIT 3.56x45 flash hider	-1.0	-6.0%	¥10,931	Peacekeeper LL3
	Magpul M-LOK 4.1 inch rail	-0.2	-	¥1,558	Mechanic LL1
	Zent RK-2 tactical foregrip	-8.0	-4.0%	¥31,908	Free Lv20
	AAC T62 SDN 6 multi-caliber sound suppressor	-26.0	-10.0%	¥47,869	Free Lv25



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Optimization Results: Max Ergo

Language

中文

选择武器

选择一把武器:

Custom Guns NL545 (...)

基础属性:

人机工效: 48

垂直后坐力: 114

水平后坐力: 330

可用提示 (1)

玩家与商人

按等级

79

高级市场

商人等级

订单条件

十一—包含特殊模组

最终属性

人机工效

100.0

↑52.8

垂直后坐力

63.7

↓38.3

水平后坐力

184.5

↓54.5

总重量

2.83 公斤

总配置费用

¥247,421

↑1494,308

已选配置

预览: Custom Guns NL545 (GP) 5.45x39 assault rifle Default

配置价格: ¥153,633 (fleaMarket)

包含: 14 items

额外模组

	名称	人机	后坐力	价格	来源
	AR-15 HK Ergo PSG-1 style pistol grip	+15.0	-	¥27,140	Mechanic LL4
	AK-74 "Saiga 545" 5.45x39 10-round magazine	+5.0	-	¥1,566	flea LL25
	AR-15 Strike Industries Advanced Receiver Extension	+2.0	-2.0%	¥10,440	Mechanic LL4
	NL545 (GP) 5.45x39 11.5 inch barrel	-6.0	-2.1%	¥25,760	Skier LL4
	HK 416 Strike Industries CRUX 15inch M-LOK handguard	+11.0	-3.0%	¥16,700	Mechanic LL4
	AR-15 FAB Defense GL-CORE buttstock	+15.0	-19.0%	¥4,715	Skier LL3
	Magpul M-LOK AFS tactical foregrip (FDG)	+7.0	-2.6%	¥8,067	Skier LL3

Custom Guns NL545 (GP) 5.45x39 assault rifle Default预览中的物品

优化详情

Optimization Results: Balanced Build

Language
中文

选择武器
选择一把武器:
Custom Guns NL545 (...)

基础属性:

- 人机工效: 48
- 垂直后坐力: 114
- 水平后坐力: 330

可用预设 (1)

玩家与商人

- 玩家等级: 79
- 跳蚤市场
- 商人等级
- 约束条件
- 包内物品删除按钮

优化温度

最终属性

人机工效 ①
71.3
+42.3

垂直后坐力 ①
54.0
-44.3

水平后坐力 ①
156.4
-173.6

总重量 ①
3.01 公斤

总配置成本 ①
¥201,889
+194,856

已选配置

预览: Custom Guns NL545 (GP) 5.45x39 assault rifle Default
套装名称: P153,033 (RealMarket)
包含: 14 items

额外模组:

	名称	人机	后坐力	价格	来源
	AR-15 Hera Arms HG-15 pistol grip	+10.0	-	¥5,100	Mechanic LL4
	AK-74 "Salvo 745" 5.45x39 10-round magazine	+5.0	-	¥1,566	Flex Lv25
	AR-15 Strike Industries Advanced Receiver Extension	+2.0	-2.0%	¥10,440	Mechanic LL4
	HK 416 Strike Industries CRUX 15 inch M-LOK handguard	+11.0	-3.0%	¥16,700	Mechanic LL4
	Magpul M-LOK 4.1 inch rail	-0.2	-	¥1,558	Mechanic LL1
	RTM Ososvets P-2 tactical foregrip (Khaki)	+1.0	-3.0%	¥13,492	Slaker LL3

Custom Guns NL545 (GP) 5.45x39 assault rifle Default预设设备中的物品

优化详情

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The Limitation of Linear Weighted Sum

Traditional Formulation: $\max W_E \cdot \text{Ergo} - W_R \cdot \text{Recoil} - W_P \cdot \text{Price}$

The Non-Convexity Problem:

- Linear weighted sum sweeps **straight hyperplanes** across the objective space.
- In Tarkov, mod compatibility creates **non-convex indentations** (e.g., discrete trade-offs between heavy suppressors vs. lightweight compensators).
- **Mathematical Blind Spot:** Any point located inside a non-convex pocket **cannot be found by linear sums** under *any* choice of weights (W_E, W_R, W_P)!

Empirical Finding (Tarkov Datasets)

Across common benchmark weapons (M4A1, AK-74N, MP7A2):

- **45%–61%** of all Pareto-optimal builds lie in non-convex pockets!
- Linear weights jump directly from extreme-recoil to extreme-ergo, missing the best **"Sweet Spot"** builds.

The 3D Ideal Reference Point

Idea: Instead of arbitrary linear slopes, measure distance to a **Utopian Ideal Target**.

1. Utopia Ideal Point (z^*):

- $z_E^* = \max_{\mathbf{x}} \text{Ergo}(\mathbf{x})$ (Best achievable Ergo)
- $z_R^* = \min_{\mathbf{x}} \text{Recoil}(\mathbf{x})$ (Best achievable Recoil)
- $z_P^* = \min_{\mathbf{x}} \text{Price}(\mathbf{x})$ (Cheapest viable build)

Computed instantly via 3 single-objective solves.

2. Normalized Gap to Target (g_i): For each objective $i \in \{E, R, P\}$:

$$g_i(\mathbf{x}) = \frac{|f_i(\mathbf{x}) - z_i^*|}{\text{Range}_i} \in [0, 1]$$

- $g_i = 0$: Reaches the theoretical best.
- Dimensionless and scale-invariant across Ergo, Recoil %, and Roubles ().

Augmented Tchebycheff Scalarization

Mathematical Formulation (Sweet Spot Mode):

$$\min_{\mathbf{x}, \alpha} \left(\alpha + \rho \sum_{i \in \{E, R, P\}} \lambda_i \cdot g_i(\mathbf{x}) \right)$$

subject to: $\lambda_i \cdot g_i(\mathbf{x}) \leq \alpha, \quad \forall i \in \{E, R, P\}$

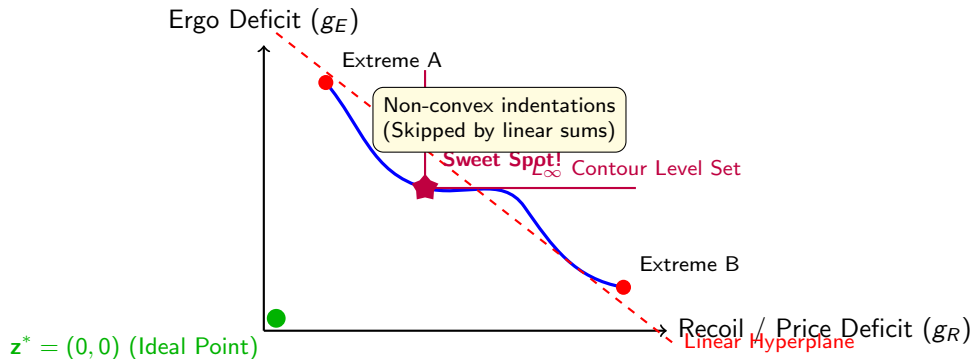
Why This Works:

- $\alpha = \max_i(\lambda_i g_i(\mathbf{x}))$ minimizes the **worst-case deficit**.
- Pulls the build uniformly towards the Ideal Point from all directions.
- Small augmentation term $\rho \sum \lambda_i g_i$ ($\rho = 10^{-4}$) guarantees **strict Pareto optimality** (eliminates flat weak optima).

MILP Integration Advantages:

- Requires only **1 extra continuous variable** (α).
- **Zero binary variables added** $\implies$ solve time remains $< 15\text{ms}$.
- Discovers non-convex sweet spots with 100% mathematical proof of optimality.

Geometric Intuition: Hyperplane vs. L_∞ Contour



Key Takeaway: Straight lines bridge across concave regions, while L-shaped Chebyshev contours penetrate deep into the "knee" to find balanced builds.

In-Game ADS Physics & The Overswing Problem

The Real-World Tarkov Combat Dilemma:

- High Ergonomics speeds up Aim-Down-Sights (ADS) transition and reduces arm stamina drain.
- However, heavy weapon weight introduces **rotational momentum and inertia**.

What is Sight Overswing?

When quickly snapping to aim, a weapon whose weight exceeds its ergonomic handling limit will **sway past the target reticle** before settling.

- Causes missed first shots in high-tempo CQB.
- A 100-ergo, 7.5 kg build overswings violently.
- A 65-ergo, 2.2 kg build snaps instantly and

"Raw Ergonomics is misleading without accounting for weapon weight."

Community Research:

SpaceMonkey37 & EFTForge established the empirical relationship between Ergonomics, Weight, and Overswing.

The Quadratic EvoErgoDelta (EED) Metric

Community Formulation (SpaceMonkey37 / EFTForge):

$$\text{EED} = [100 - 0.015 \cdot (100 - \text{Ergo})^2] - 15 \cdot \text{Weight}_{\text{kg}}$$

The Physical Meaning of EED:

- **EED ≥ 0 (Zero Overswing):** Weight is within the ergonomic threshold; sights snap cleanly on target.
- **EED < 0 (Overswing Detected):** Weapon inertia causes crosshair sway upon entering ADS.

Linearized Approximation (EvoErgo):

$$\text{EvoErgo} = \min(100, \text{Ergo}) - 15 \cdot \text{Weight}_{\text{kg}}$$

- Standard community linear score displayed in results overview cards.
- Balances ergonomics and weight directly at a 1:15 kg exchange rate.

Solving Non-Linear EED: The Tangent k -Sweep Solver

Optimization Challenge: EED is quadratic and non-linear, but LP/MIP solvers require linear objectives for sub-millisecond execution.

Tangent Exchange Rate: The marginal value of ergonomics varies with current ergo E :

$$k(E) = \frac{\partial \text{EED}}{\partial E} = 0.03 \cdot (100 - E) + 15 \quad (\text{Exchange slope } \Delta W / \Delta E)$$

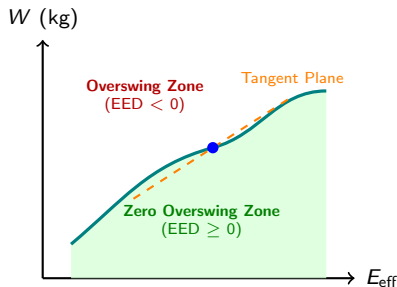
The Tangent k -Sweep Algorithm (`solveEvoErgo`)

- 1 Run candidate linear solves across benchmark exchange rates $k \in \{15.0, 10.0, 7.5, 5.6\}$ and $k(E_0)$.
- 2 Obtain integer-feasible candidate loadouts $\mathbf{x}^{(1)}, \mathbf{x}^{(2)}, \dots, \mathbf{x}^{(m)}$.
- 3 Evaluate exact quadratic $\text{EED}(\mathbf{x}^{(j)})$ on each candidate.
- 4 Return the global maximum $\mathbf{x}^* = \arg \max \text{EED}(\mathbf{x}^{(j)})$ in $< 10\text{ms}$.

Hard Constraint: Prevent Sight Overswing

Requirement: Ensure the optimized weapon guarantees **Zero Overswing** ($EED \geq 0$).

$$\text{Weight}_{\text{kg}} \leq \text{Threshold}(E_{\text{eff}}) = \frac{100 - 0.015 \cdot (100 - E_{\text{eff}})^2}{15}$$



Tangent Cutting Planes:

- Replaces the quadratic curve with first-order tangent cuts:

$$W \leq T(E_0) + T'(E_0)(E - E_0)$$

- Convex upper bound: zero binary variables needed, solving in $< 15\text{ms}$.

Equipment Ergonomics Modifier (b)

Real Combat Scenario: Body armor, armored rigs, helmets, and backpacks reduce your effective ergonomics.

Effective Ergonomics Loadout Formula

$$E_{\text{eff}} = E_{\text{weapon}} \cdot (1 + b), \quad b \in [-0.40, 0.00]$$

- b : Total equipment ergo penalty percentage from player loadout.
- Example: A heavy Tier-6 Zabralo armor ($b = -28\%$) + Altyr helmet ($b = -7\%$)
 $\implies b = -35\%$.

Player Benefit: The optimizer automatically adjusts the overswing threshold to account for your armor, guaranteeing that the build snaps with zero sway in full raid gear.

Problem: Gunsmith quests require building weapons to exact specifications.

Task Definition (JSON):

- Target weapon name
- Stat constraints:
 - Min ergonomics, max recoil
 - Min magazine capacity
 - Max weight, min sighting range
- Required specific items
- Required categories (e.g., “Silencer”)

How It Works:

- 1 Load task from `tasks.json`
- 2 Convert requirements to constraints:
 - Required items $\rightarrow X_i = 1$
 - Categories $\rightarrow \sum_{i \in \text{cat}} X_i \geq 1$
 - Stats $\rightarrow$ hard bounds
- 3 Objective: minimize price
- 4 Solver finds cheapest valid build

Result: Automatically solves all 25 Gunsmith quests with optimal (cheapest) builds.

Gunsmith Task List

机械师任务优化器

选择任务

Gunsmith - Part 6 (AKM)

任务要求

武器: Kalashnikov AKM 7.62x39 assault rifle

- 最低人机工效: 40
- 最大后坐力总和: 400
- 最小瞄准距离: 800

所需物品:

- Fortis Shift tactical foregrip
- AK 7.62x39 Magpul PMAG 30 GEN M3 30-round magazine
- One of: Silencer

按照最低价格排序

✓ 最优

✓ 优化 最优

最终属性

人机工效 ⓘ	垂直后坐力 ⓘ	水平后坐力 ⓘ	总重量 ⓘ	总配置费用 ⓘ
40.0	110.3	279.7	5.49 公斤	¥124,564
↑ 14.0	↓ 38.7	↓ 98.3		↑ 130,564

Gunsmith Solution Details

已选配置

预览: Kalashnikov AKM 7.62x39 assault rifle (AKMP)

套装价格: ¥35,000 (fleaMarket)

包含: 11 Items



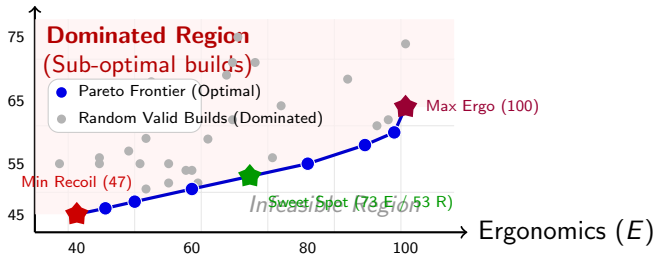
额外配件:

	名称	人机	后坐力	价格	来源
	AKM Hexagon 7.62x39 sound suppressor	-23.0	-6.0%	¥25,223	Flea Lv25
	AK Aeroknox Scorpius pistol grip	+12.0	-	¥5,800	Mechanic LL4
	AK Akademia Bastion dust cover	+1.0	-0.5%	¥6,325	Skier LL1
	AK Samson Rear Trunnion Folding Stock Adapter	-0.5	-	¥1,093	Mechanic LL3
	AK 7.62x39 Magpul PMAG 30 GEN M3 30-round magazine	-1.5	-	¥5,513	Peacekeeper LL2
	AK 100-series polymer handguard	+5.0	-	¥1,441	Prapor LL1
	KMZ 1P69 Weaver mount	-1.0	-	¥2,050	Prapor LL2
	SIG Sauer Collapsing/Telescoping Stock	+17.0	-18.0%	¥4,315	Peacekeeper LL1
	Fortis Shift tactical foregrip	+8.0	-1.5%	¥21,496	Flea Lv20
	KMZ 1P59 3-10x riflescope	-8.0	-	¥13,680	Flea Lv25
	KMZ 1P59 scope eyecup	+5.0	-	¥628	Prapor LL2

Pareto Frontier Analysis: Real Weapon Benchmark (M4A1)

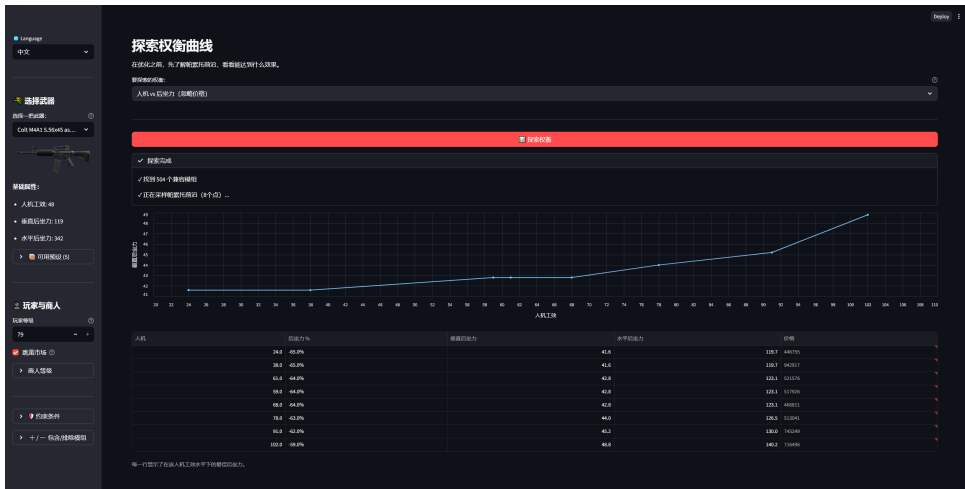
Core Concept: The Pareto Frontier represents the boundary of non-dominated builds where no stat can be improved without sacrificing another.

Vertical Recoil (R , lower is better)



Takeaway: Points inside the cloud represent wasted performance (e.g. higher recoil for the same ergonomics). The optimizer mathematically guarantees you stay on the blue boundary.

Pareto Frontier Visualization



Future Work & Unimplemented Features

- **Hidden Weapon Stats:** Currently unable to retrieve hidden weapon stats, such as recoil angle.
- **Trader Barters:** Currently unable to retrieve trader barter items and their calculated prices.

Data Source



Open-source GraphQL API for Escape from Tarkov

AI Coding Assistants



Claude Code
Anthropic



Gemini CLI
Google



MiniMax
MiniMax

All the AI assistants helped me with code generation, debugging, and documentation